12B — Deep-Dive Feature Audit: UI, Rendering, Serialization & Master Architecture (Part 2 of 4)

Canvas 2D Rendering (`js/render.js`), The `31 REL_V230_*` Relic Aura Black Hole, Visual-to-Lore Fidelity (`Bio-Age`, `Qinggong`, `Peak Lv.3–5` Sprites), Item Art Parity & `Zero-RNG` Purity *(Audio Excluded per Project Scope)*

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Primary Audited Modules**: `js/render.js`, `js/skills.js` (`VFX registries`), `js/ui.js` (`ITEM_ART_CATALOG`), `js/world_map.js`
- **Explicit Scope Exclusion**: **Audio / Sound (SFX) is intentionally excluded** per project direction (เกมส์ไม่เน้นเสียง). Do not spend engineering cycles or payload bytes expanding audio synthesis.
- **Immutable Visual Rules**:
    1. **100% Pure Read-Only Rendering**: `R.draw`, `Render.portrait`, `Render.diorama`, `Render.buildingShowcase`, and `UI.renderAtlasIconHtml` must **NEVER** advance `W.rng` or mutate `SIM.serialize(W)`.
    2. **Balanced Canvas Context**: Every path through `render.js` must guarantee `ctx.save()` === `ctx.restore()` depth (`0` net leak) and finite numeric coordinates.
    3. **Preserve Existing Visual Suite Counts**: Maintain `45 SIGNATURE_VFX_CATALOG`, `50 T6_VFX_FAMILIES`, `>= 60` forms per tier `T1–T7` (`v260_vfx_tier_subpath_equip_suite.js`), `27 VC` relics in `LEGENDARY_GEAR_VISUALS`, and `ap.hasSilkRibbon === true` (`v250:166`).

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1. [REN-LEG-01] Critical Visual Black Hole on All 31 Grand Dao Divine Relics (REL_V230_<PATH>) & 2 Capstone Arts (LEG_V240_GOD / CRAFT_CREATION)!

1.1 Exact Proof From legendary_visual_identity_suite.js:22 vs t6_group_vfx_suite.js:68

Trace what happens when a Stage-8 Grandmaster equips one of the **31 Grand Dao Divine Relics (`REL_V230_<PATH>`, `tier: 7, legendary: true`)**:

1. `R.LEGENDARY_GEAR_VISUALS` contains **only `27` entries** (`the 20 weapons, 3 armors, and 4 accessories from VC.LEGENDARY_EQUIPMENT`). Thus `R.resolveLegendaryGearVisual(relV230Item)` returns **`null`**—so **`drawLegendaryEquipmentAura` and `drawLegendaryWeaponVisual` draw NOTHING**!
2. Next, `drawT7EquipmentAura` (`verified in t6_group_vfx_suite.js:68`) explicitly skips any item with **`it.legendary === true`** (`assuming drawLegendaryEquipmentAura handled it`)!
3. **Paradoxical Result**: An ordinary non-legendary Tier-7 iron sword displays a radiant T7 Equipment Aura, **whereas a Mythic Grand Dao Divine Relic (`REL_V230_*`, `legendary: true`) falls into the crack between both renderers and displays `0` equipment aura and `0` custom weapon silhouette**!

1.2 Test-Safe Remediation Blueprint (Preserving Object.keys(R.LEGENDARY_GEAR_VISUALS).length === 27!)

- Note `legendary_visual_identity_suite.js:22` asserts `assert.equal(Object.keys(R.LEGENDARY_GEAR_VISUALS).length, 27)`. **Do NOT mutate the keys count of `R.LEGENDARY_GEAR_VISUALS`!**
- Instead, add a separate **`R.GRAND_DAO_RELIC_VISUALS`** map (`31` entries keyed `REL_V230_<PATH>`, deriving palette & crown glyph from `R.PATH_VISUAL_SIGNATURES[path]`) consulted inside `R.resolveLegendaryGearVisual(it)` (`after checking` R.LEGENDARY_GEAR_VISUALS[it.id]`, or as a dedicated` drawGrandDaoRelicAura`branch when`it.legendary && !R.LEGENDARY_GEAR_VISUALS[it.id]`)!
- Similarly, `legendary_visual_identity_suite.js:21` asserts `Object.keys(R.LEGENDARY_ART_VISUALS).length === 50`. Route **`LEG_V240_GOD` and `LEG_V240_CRAFT_CREATION` (`it.apexPathArt === true`)** into a dedicated `R.APEX_PATH_ART_VISUALS` fallback inside `drawCastEffects` so

GOD and CRAFT_CREATION capstone arts receive full 60–84 px Ascendant Apex Apparitions instead of ordinary T7 rings!

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2. [REN-CTX-02] Visual-to-Lore Fidelity Gaps Across Characters, Movement & Sect Architecture (js/render.js)!

1. **Chronological Age (ageYears) vs Biological Age (HC.bioAge) in Hair/Face Aging**:
   - In Xianxia lore, a 90-year-old Stage-7 Nascent Soul Immortal (lifespanMax = 750, bioAge = 9.6 — youthful prime!) retains lustrous dark hair and smooth features, whereas a 68-year-old Rank-0 mortal steward (lifespanMax = 80, bioAge = 68) grows silver hair and elder lines!
   - Ensure r.appearance(n) drives visual aging cues strictly off **HC.bioAge(n)** (never raw ageYears), **and** includes n.h.body.hairHex, n.h.body.marks.length, and Math.floor(HC.bioAge(n) / 10) in the apCache signature so TR_AC_HEART_DEMON_LOVE overnight white-hair transformation (11A) and battle scars render immediately!
2. **Path-Authentic Apex Regalia Styles (Keeping ap.hasSilkRibbon === trueforv250:166!)**:
   - Keep ap.hasSilkRibbon = true at isPathApex so v250 passes 100%, while selecting the **visual silhouette** of that ribbon/mantle by sig.family:
     - Daoist/Celestial/Scholar/Music paths $\to$ Classic floating Celestial Silk Ribbon (ผ้าแพรเซียนลอยฟ้า).
     - SHADOW_SAMURAI / NINPO_* $\to$ Flowing Crimson/Shadow Haori War-Tassels (พู่เสื้อคลุมซามูไร/ผ้าพันคอนินจา).
     - BODY / ARENA_GLORY / DEVOURING_FLESH $\to$ Blazing Blood-Qi Battle Mantle (รัศมีปราณโลหิตศึก).
     - BUDDHA_DHARMA / KARMA_SAMSARA / OCCULT_MAGE $\to$ Sacred Kasaya Dharma Stole & Prayer Beads (จีวรแสงธรรม/สายยันต์อาคม).
3. **Path-Authentic QINGGONG Airborne Visuals (03D Cross-Link)**:
   - When SHADOW_SAMURAI, BODY, or ARENA_GLORY moves at high tier (GALE_STRIDE / SKY_STEP), draw **wind-step afterimages / shockwave rings** underfoot rather than placing a Zero-Qi Samurai atop a Daoist flying jian or fairy cloud!
4. **Unlocking Unseen Lv.3–5 Peak Superstructures & Plotting the 3 Virtual Facilities (07B / 07C)**:
   - Render.buildBuildingSprite (v4.1.0) already contains complete, tested Stage-2 (Lv.3–4 vermilion Dou-Gong) and Stage-3 (Lv.5+ white-jade dragon pillars & floating runes) artwork for all 8 Peak Sanctuaries—fixing SECT.upgradeBuilding (07B) immediately brings this hidden artwork to life in natural runs! Likewise, placing mech_workshop, beast_menagerie, and

`harmonic_pavilion` onto `MAP.PLOTS` (`07C`) makes all 3 specialist halls visible on the mountain!

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3. `[REN-VFX-03]` Remaining `castFx` & Status-Aura Visual Edge Cases (`js/combat.js` $\leftrightarrow$ `js/render.js`)!

1. **Channeled & Multi-Tick Skills (`C.resolveChannelTick`, `combat.js:1044`) Emit Zero Follow-Up `castFx` Pulses**:
   - `useSkill` emits `a.castFx` once on Tick 0, **but `C.resolveChannelTick` (`Ticks 1..N`, including the massive `1.65× isFinal` release of `startChargedSkill`!) NEVER emits `a.castFx`**!
   - **Visual Consequence**: During a 2-tick Charged Ultimate, the big slash VFX plays prematurely at the *start* of charging (`Tick 0`), and when the actual `1.65×` Charged Slash lands 2 ticks later, **nothing plays on screen**!
   - **Fix**: For `isChargedSkill`, emit a compact charge-gathering ring on Tick 0 and emit the full `signatureVfx` / `tierVfx` release `castFx` (`seq++`) on the `isFinal` tick inside `C.resolveChannelTick`!
2. **Synchronize `ITEM_ART_CATALOG.typeAliases` Back Into `MARTIAL.gearType` (`05A` Cross-Link)**:
   - `equipment_tier_art_suite.js:20` proves `UI.ITEM_ART_CATALOG.typeAliases` maps `HEAVY_ARMOR -> HEAVY`, `LIGHT_ARMOR -> LIGHT`, `HEADGEAR -> HELMET`, `SHOES -> BOOTS`, `SABER -> BLADE`, `GAUNTLET -> FIST`. Sharing that exact alias normalization with `MARTIAL.gearType` ensures gameplay mechanics always match the exact item icon shown to the player!
3. **Bounded `apCache` Eviction (`Memory Hygiene`)**:
   - Prune `apCache` entries for IDs no longer in `W.byId` (`buried/retired NPCs`) whenever `apCache.size > 256` so 300-year runs never leak cached appearance objects.